

Lecture 6 – Point Estimation

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Outline:

6.1	Point Estimation	1
6.1.1	Method of Moments	1
6.1.2	Method of Maximum Likelihood	3

6.1 Point Estimation

- **Definition:** Given a statistical model $\{f_\theta(x) \mid \theta \in \Theta\}$, any statistic $T = T(\underline{X})$ used for estimating $g(\theta)$ is called a **point estimator**.

- Estimator – refers to the *function*, i.e., the random variable $T(\underline{X})$
- Estimate – refers to a *particular value*, i.e., the realization $T(\underline{x})$

- In this lecture, we address two basic questions:

1. How do we find estimators?
2. How do we compare / evaluate different estimators?

6.1.1 Method of Moments

- For a sample $X_i \stackrel{\text{iid}}{\sim} f_\theta(x)$ for $\theta \in \Theta \subset \mathbb{R}^k$ we can define

- **Theoretical moments:**

$$\mu_k = \mathbb{E}[X_i^k] \quad (\text{depend on } \theta)$$

- **Empirical moments:**

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k \quad (\text{are statistics})$$

- **Intuition:** A good estimation procedure should lead to a match between theoretical and empirical moments.
- In order to estimate θ , the method of moments requires to solve

$$\begin{aligned} \mu_1(\theta_1, \dots, \theta_k) &= m_1 \\ \mu_2(\theta_1, \dots, \theta_k) &= m_2 \\ &\vdots \\ \mu_k(\theta_1, \dots, \theta_k) &= m_k \end{aligned}$$

This is a system of k equations with k unknowns.

- **Example:** $X_i \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$. Find the method of moments estimator of θ .

- In this case, $k = 1$ and theoretical first moment is

$$\mu_1 = \mathbb{E}[X_i] = \frac{1}{\theta}$$

- The empirical first moment is

$$m_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$$

- Equating these, we see that

$$\mu_1 = m_1 \quad \Longleftrightarrow \quad \frac{1}{\theta} = \bar{X}_n \quad \Longleftrightarrow \quad \theta = \frac{1}{\bar{X}_n}$$

- Hence, $\hat{\theta} = \frac{1}{\bar{X}_n}$ is the method of moments estimator.

- **Example:** $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$

- $k = 2$ and the theoretical moments are

$$\mu_1 = \mathbb{E}[X_i] = \mu, \quad \mu_2 = \mathbb{E}[X_i^2] = \mathbb{E}^2[X_i] + \text{Var}(X_i) = \mu^2 + \sigma^2$$

- Matching the theoretical and empirical moments gives:

$$\begin{aligned} \mu_1(\mu, \sigma^2) = m_1 \\ \mu_2(\mu, \sigma^2) = m_2 \end{aligned} \quad \Longleftrightarrow \quad \begin{aligned} \mu &= m_1 \\ \mu^2 + \sigma^2 &= m_2 \end{aligned} \quad \Longleftrightarrow \quad \begin{aligned} \mu &= m_1 \\ \sigma^2 &= m_2 - m_1^2 \end{aligned}$$

- Thus, the moment estimators are

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n, \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2$$

- Letting $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ denote the empirical mean, the estimator of the variance can also be expressed in terms of the empirical variance:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Hence, the moment estimators for the Normal model are exactly the empirical mean and variance.

- **Remark:** There is some arbitrariness in the method of moments. For example, why match the first k moments but not others? Why not match moments or some function g of the observe data?

- **Example:** $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, $\theta = \sigma^2$.

- In this case, $\mu_1(\theta) = \mathbb{E}[X_1] = 0$ for all θ and thus “solving” the equation $\mu_1(\theta) = m_1$ is not possible in general.

- As an alternative, one may consider the second moment:

$$\mu_2(\theta) = \mathbb{E}[X_i^2] = \sigma^2 \quad \Longleftrightarrow \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

- But, one could also have considered matching the fourth moment:

$$\mu_4 = \mathbb{E}[X_1^4] = 3\sigma^4 \quad \Longleftrightarrow \quad \hat{\sigma}^2 = \left(\frac{1}{3n} \sum_{i=1}^n X_i^4 \right)^{1/2}$$

- **Example:** $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$.

- $k = 1$ and the first theoretical moment is

$$\mu_1 = \mathbb{E}[X_i] = \int_0^\theta x \frac{1}{\theta} dx = \frac{\theta}{2}$$

- Solving for θ such that $\mu_1(\theta) = m_1$ gives the moment estimator:

$$\hat{\theta} = 2\bar{X}_n$$

But does this estimator make sense?

- Note that this estimator is not a function of the sufficient statistic $X_{(n)}$. So in a sense it is not using all the relevant information.
- If $n = 3$ and $\underline{x} = (1, 24, 2)$ then $\hat{\theta} = 18$. However, we have observed 24 so we know that $\theta \geq 24$.

6.1.2 Method of Maximum Likelihood

- **Definition:** Given a statistical model $\{f_\theta(x) \mid \theta \in \Theta\}$, assuming $\underline{X} = \underline{x}$ is observed, recall from the Lecture 3-4 notes (Section 3.6) that the **likelihood function** is given by

$$L_{\underline{x}}(\theta) = f_{\underline{X}, \theta}(\underline{x}), \quad \text{for } \theta \in \Theta$$

As in the previous lectures, the subscript $\underline{x}$ records the fixed observed sample, while θ is the variable of the likelihood function.

- The likelihood function looks at the joint distribution of the sample in a different way:

- $f_{\underline{X}, \theta}(\underline{x}) = \prod_{i=1}^n f_\theta(x_i)$ is a function of $\underline{x}$ for fixed θ :
- $L_{\underline{x}}(\theta)$ is a function of θ for fixed $\underline{x}$

- In the discrete case:

- The likelihood is the probability of observing $\underline{x}$ when θ is the true parameter, i.e.,

$$L_{\underline{x}}(\theta) = f_{\underline{X}, \theta}(\underline{x}) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

- Comparing probabilities, we see that

$$\frac{L_{\underline{x}}(\theta_1)}{L_{\underline{x}}(\theta_2)} > 1 \quad \Longrightarrow \quad \text{the probability of observing } \underline{x} \text{ is larger if } \theta = \theta_1 \text{ than if } \theta = \theta_2$$

Sometime we say that θ_1 is *more likely* than θ_2 .

- **Remark:** The statement *more likely* is not the same as the statement *more probable*. Note that θ is not a random variable so there is no probability to assign to these events. See https://en.wikipedia.org/wiki/Likelihood_function#Historical_remarks for more information on the history of this terminology.

- **Intuition:** The likelihood function $L_{\underline{x}}(\theta)$ induces an **ordering** among the θ . We prefer values of θ which make the realization $\underline{x}$ more probable. Thus, we only care about the ordering, not the precise value of L .
- **Likelihood principle:** If $\underline{x}$ and $\underline{x}^*$ are such that $L_{\underline{x}}(\theta)$ is proportional to $L_{\underline{x}^*}(\theta)$, i.e. there exists a constant $c(\underline{x}, \underline{x}^*)$ such that

$$L_{\underline{x}}(\theta) = c(\underline{x}, \underline{x}^*) L_{\underline{x}^*}(\theta), \quad \text{for all } \theta \in \Theta,$$

then inference based on $\underline{x}$ and $\underline{x}^*$ should coincide. Note that proportional likelihoods lead to the same ordering on θ .

- **Example.** Consider $X_i \stackrel{i.i.d.}{\sim} \text{Poisson}(\theta)$. Suppose $\underline{x} = (1, 2)$ and $\underline{x}^* = (0, 3)$. Then

$$L_{\underline{x}}(\theta) = e^{-2\theta} \frac{\theta^3}{1!2!}, \quad \text{and} \quad L_{\underline{x}^*}(\theta) = e^{-2\theta} \frac{\theta^3}{0!3!}.$$

So $L_{\underline{x}}(\theta) = 3L_{\underline{x}^*}(\theta)$, and thus the likelihood principle implies that the inference based on $\underline{x}$ and $\underline{x}^*$ should coincide.

- **Definition:** Given a statistical model $\{f_{\theta}(x) \mid \theta \in \Theta\}$ and random sample $\underline{X}$ with realization $\underline{x}$ the **maximum likelihood** estimate is $\hat{\theta} = \hat{\theta}(\underline{x})$ such that

$$L_{\underline{x}}(\hat{\theta}) = \max_{\theta \in \Theta} L_{\underline{x}}(\theta)$$

Then, the **Maximum Likelihood Estimator** (MLE) is $\hat{\theta} = \hat{\theta}(\underline{X})$. Note that this is a random variable because it is a function of the sample.

- **Finding the MLE:** Assume that $\Theta \subset \mathbb{R}$ and the likelihood function is smooth.
 - If the likelihood has a unique maximizer $\hat{\theta}$ in the interior of Θ (i.e. not at the boundary), then it satisfies the derivative conditions:

$$\left. \frac{\partial}{\partial \theta} L_{\underline{x}}(\theta) \right|_{\theta=\hat{\theta}} = 0, \quad \text{and} \quad \left. \frac{\partial^2}{\partial \theta^2} L_{\underline{x}}(\theta) \right|_{\theta=\hat{\theta}} < 0$$

- It is often convenient to work with the **log-likelihood function**:

$$\ell_{\underline{x}}(\theta) = \log L_{\underline{x}}(\theta)$$

- Because the logarithm is strictly increasing, it does not affect the location of the maximum, and the derivative conditions can be stated equivalently as

$$\left. \frac{\partial}{\partial \theta} \ell_{\underline{x}}(\theta) \right|_{\theta=\hat{\theta}} = 0, \quad \text{and} \quad \left. \frac{\partial^2}{\partial \theta^2} \ell_{\underline{x}}(\theta) \right|_{\theta=\hat{\theta}} < 0$$

- **Example:** $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, $\Theta = [0, 1]$

- The likelihood is

$$L_{\underline{x}}(\theta) = \prod_{i=1}^n f_{\theta}(x_i) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}.$$

- First we consider the case $\sum_{i=1}^n x_i = n$ (all 1's). The likelihood is

$$L_{(1,\dots,1)}(\theta) = \theta^n$$

which is increasing in θ . Thus, the maximum is achieved at $\hat{\theta} = 1$.

- Next, we consider the case $\sum_{i=1}^n x_i = 0$ (all 0's). The likelihood is

$$L_{(0,\dots,0)}(\theta) = (1 - \theta)^n$$

which is decreasing in θ . Thus, the maximum is achieved at $\hat{\theta} = 0$.

- Finally, we consider the case $0 < \sum_{i=1}^n x_i < n$. Noting that

$$L_{\underline{x}}(0) = L_{\underline{x}}(1) = 0 \quad \text{and} \quad L_{\underline{x}}(\theta) > 0 \quad \text{for } 0 < \theta < 1$$

we conclude that the maximum does not occur at the boundary points and we can restrict our attention to the interval $\theta \in (0, 1)$.

- To find the maximizer, it is convenient to work with the logarithm of the likelihood function (a.k.a. the log likelihood), which is given by

$$\ell_{\underline{x}}(\theta) = \sum_{i=1}^n \log f_{\theta}(x_i) = \sum_{i=1}^n x_i \log(\theta) + \left(n - \sum_{i=1}^n x_i\right) \log(1 - \theta).$$

- This function is differentiable in $\theta \in (0, 1)$ with

$$\begin{aligned} \frac{\partial}{\partial \theta} \ell_{\underline{x}}(\theta) &= \frac{1}{\theta} \sum_{i=1}^n x_i - \frac{1}{1 - \theta} \left(n - \sum_{i=1}^n x_i\right) \\ &= \frac{1}{\theta(1 - \theta)} \left[(1 - \theta) \sum_{i=1}^n x_i - \theta \left(n - \sum_{i=1}^n x_i\right) \right] \\ &= \frac{n}{\theta(1 - \theta)} \left[\frac{1}{n} \sum_{i=1}^n x_i - \theta \right] \end{aligned}$$

- The derivative has a unique root at $\theta = \bar{x}_n$. Since the derivative is strictly positive for $\theta < \bar{x}_n$ and strictly negative for $\theta > \bar{x}_n$ we conclude that this root is the unique global maximum, and thus the maximum likelihood estimate is $\hat{\theta} = \bar{x}_n$.
- Note that an alternative approach to verifying that the root of the derivative corresponds to a maximum is to consider the second derivative;

$$\frac{\partial^2}{\partial \theta^2} \ell_{\underline{x}}(\theta) = -\frac{1}{\theta^2} \sum_{i=1}^n x_i - \frac{1}{(1 - \theta)^2} \left(n - \sum_{i=1}^n x_i\right) < 0, \quad \text{for all } \theta,$$

Since the second derivative is strictly negative for all θ , we conclude that the root corresponds to the unique global maximum.

- Combining all three cases, we see that the MLE is given by the sample mean:

$$\hat{\theta} = \bar{X}_n \quad \text{is the MLE}$$

- **Example:** $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $\Theta = \mathbb{R}_+$

- The likelihood is

$$L_{\underline{x}}(\theta) = \frac{1}{\theta^n} \mathbf{1}_{x_{(n)} \leq \theta}$$

- This function is not differentiable in θ because there is a discontinuity at $\theta = x_{(n)}$.
- However the maximum can be determined by inspection:
 1. $L_{\underline{x}}(\theta)$ is zero for $\theta < x_{(n)}$
 2. $L_{\underline{x}}(\theta)$ is strictly positive and decreasing in θ for $\theta \geq x_{(n)}$

Thus, the unique maximum $\hat{\theta} = X_{(n)}$ is the MLE.